# GESTURE-CONTROLLED PHONE LIGHT

A webcam, hand tracking, and your Android phone become a small interactive IoT project.

### Project at a glance

Show a hand to the computer webcam. Pinch thumb and index finger to request flashlight OFF; spread them apart to request ON. Each state change also requests a phone vibration through nikki.build Streams.

**Verified in a live run: the phone responded to both gestures with the expected flashlight change and vibration. A direct Streams ON/OFF test also worked.**

## 1. What you need

- Android phone with nikki.build Streams and an active joined session.
- Computer with webcam, Windows PowerShell, and Python 3.10 or newer (64-bit recommended).
- Phone and computer on the same local network; this project uses 10.172.21.252:3000.
- The project folder or downloadable ZIP, which includes the MediaPipe hand model.

## 2. Install and run on Windows

Unzip the project, open PowerShell in the extracted folder, and start the launcher:

```
Set-ExecutionPolicy -Scope Process Bypass
.\run.ps1
```

The launcher prepares a virtual environment, installs pinned packages, and starts the program. If dependencies are already installed, run:

```
.venv\Scripts\python.exe .\main.py
```

At the hidden Streams Session ID prompt, enter the ID for the session joined on your phone. The app does not echo or save it.

## 3. Connect the phone to Streams

- Open Streams on Android and join the active session.
- Make sure flashlight and haptics are enabled and permitted in Streams and Android.
- Confirm the socket address and session ID match the joined phone session.
- The Python SDK expects host:port without ws://; it builds the WebSocket URL itself.

# 4. Use the gestures

| Gesture | Normalized distance | Phone action |
| --- | --- | --- |
| Pinch thumb and index finger together | 0.34 or less | Send flashlight OFF plus haptic request |
| Spread thumb and index finger apart | 0.55 or more | Send flashlight ON plus haptic request |

- Hold one hand about 30-50 cm from the webcam; keep the whole hand and all fingertips visible.
- Face your palm toward the camera and use bright, even lighting.
- Hold each gesture steady for four frames. A short neutral gap between pinch and spread helps distinguish changes.
- Press Q or Esc while the preview window is active to quit. The phone remains in its last commanded state.

# 5. How it works

### Capture

OpenCV reads webcam frames and mirrors the preview.

### Track

MediaPipe Tasks estimates 21 hand landmarks.

### Measure

The program divides thumb-tip to index-tip distance by palm width, normalizing for hand size and distance.

### Filter

Pinch/spread thresholds and four stable frames reduce accidental changes.

### Control

The Streams SDK sends a light status message followed by a haptic request over WebSocket.

**Streams sends commands such as {"light": true} and {"haptic": true, "duration": 1000}. The SDK does not acknowledge phone-side execution, so its send log means a message was sent; verify the result on the phone.**

# 6. Test Streams without the webcam

The optional direct check separates phone connectivity from camera tracking. Run it from the project folder and watch the phone:

```
.venv\Scripts\python.exe .\streams_smoke_test.py
```

It sends ON plus a 500 ms haptic request, waits three seconds, then sends OFF plus haptic. Enter the same hidden Session ID.

# 7. Troubleshooting

### Preview says Show one hand

Make sure the preview shows a live image. Move the full hand farther from the lens, face the palm toward the webcam, and improve lighting.

### Black image or crossed-out camera symbol

Open the webcam privacy shutter and allow desktop apps under Windows Settings > Privacy & security > Camera. Restart the project.

### Streams cannot connect

Confirm the phone and PC share a network, Streams is open and joined, and the socket address and session ID match. The terminal reports a successful connection.

### Phone does not react

The SDK does not confirm phone-side execution. Check phone flashlight/haptics permissions and the joined session. Run streams_smoke_test.py while watching the phone. If the direct check works, restart the gesture app and retry.

### Camera unavailable

Close other camera apps. To try another webcam, set CAMERA_INDEX before launch, for example $env:CAMERA_INDEX = "1".

# 8. Project files

| File | Purpose |
|---|---|
| main.py | Gesture tracking and Streams control loop |
| streams_smoke_test.py | Direct flashlight and haptic diagnostic |
| requirements.txt | Pinned Python package versions |
| models/hand_landmarker.task | Bundled MediaPipe hand model |
| run.ps1 | Windows setup and launcher |
| README.md | Text setup and project notes |
| MODEL_LICENSE.txt | Model source and license |

# References

nikki.build Streams with Python guide supplied with the project; official SDK repository: github.com/nikki-build/streams.python. MediaPipe Hand Landmarker documentation: ai.google.dev/edge/mediapipe/solutions/vision/hand_landmarker/python.

The phone session ID is deliberately omitted. Enter it locally at runtime and do not include it in shared copies.